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Enabling scalable Model Predictive Control design for building HVAC systems using semantic data modelling

Lu Wan^{a,b}, Ferdinand Rossa^a, Torsten Welfonder^a, Ekaterina Petrova^b, Pieter Pauwels^b

^a Advanced Energy Systems, Robert Bosch GmbH, 71272 Renningen, Germany

^b Department of the Built Environment, Eindhoven University of Technology, 5600 MB Eindhoven, Netherlands

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ABSTRACT

Model Predictive Control (MPC) is a promising optimal control technique to reduce the energy consumption of Heating, Ventilation, and Air Conditioning systems in buildings. However, MPC currently involves significant manual efforts in data preparation, control model design, and software interface design. Better semantic representations of buildings, their systems, and telemetry data could help address these challenges. This paper proposes a standard semantic information model and tooling, tailored to BIM software, to streamline MPC design. The approach is tested in an office building, and the generated semantic graph is validated against a use case, where an MPC controller uses Resistance and Capacitance (RC) models that need to be parameterized. The results show that the automatically identified RC models achieve three-hour-ahead temperature predictions for two different rooms within 0.3 °C accuracy. This indicates that semantic data modelling can enable a scalable MPC configuration workflow and more efficient algorithm development and deployment in the future.

1. Introduction

Heating, Ventilation, and Air-Conditioning (HVAC) systems take up more than 50% of the energy demand in buildings [1] to preserve thermal comfort and hygienic conditions. Nevertheless, non-negligible energy waste is situated in the control of the building energy systems. More specifically, the control system parameters are often poorly tuned, and building energy systems lack multi-objective control [2]. Studies suggest that implementing an advanced optimal control scheme to the HVAC system mitigates the problem and enables great energy-saving potentials [3]. This can be achieved using Model Predictive Control (MPC) algorithms.

1.1. Model predictive control

MPC is “a class of control strategies which uses optimization to calculate actuated values in each sampling step, consisting of a cost function and constraints” [4]. In the specific case of buildings, MPC allows to represent the dynamics of a building energy system and determines its optimal operation by solving a constrained Optimal Control Problem (OCP) recursively, within a defined prediction horizon [5]. Multiple studies in the past decade have proven that MPC brings prominent benefits in demand response [6], occupant-centric control [7], and improvement of energy system efficiency [8]. In real-life MPC experiments, around

15%–50% HVAC energy consumption [9–12] and 10%–40% HVAC cost [13–15] have been reduced over weeks, demonstrating promising performance. However, the number of field implementations in the building sector is still quite limited compared with the simulation studies. This is due to the complexity of MPC applications in existing buildings.

As displayed in Fig. 1, the typical design process of a real-life MPC application follows three main stages: the data preparation (Stage 1), MPC algorithm design (Stage 2), and deployment of the proposed MPC scheme (Stage 3). The first stage deals with collecting raw data and pre-processing it properly for MPC design. Raw data can be roughly categorized into three kinds: metadata of the building and its energy systems, historical measurements of sensors, and real-time data for closed-loop control. Metadata usually originates from the design plan, represented in spreadsheets, design documentation, or nowadays more often by a Building Information Model (BIM) [16]. Historical measurements are often stored as records in a vendor-specific time series database (DB), while real-time stream data comes from the Internet of Things (IoT) platform. As the raw data originate from diverse proprietary sources, domain knowledge about such as the architecture and mechanical systems is often required to extract and prepare such data manually. In the second stage, the control engineers select a

* Corresponding author at: Advanced Energy Systems, Robert Bosch GmbH, 71272 Renningen, Germany.

E-mail addresses: lu.wan@de.bosch.com (L. Wan), f.rossa@ib-rossa.de (F. Rossa), torsten.welfonder@de.bosch.com (T. Welfonder), e.petrova@tue.nl (E. Petrova), p.pauwels@tue.nl (P. Pauwels).

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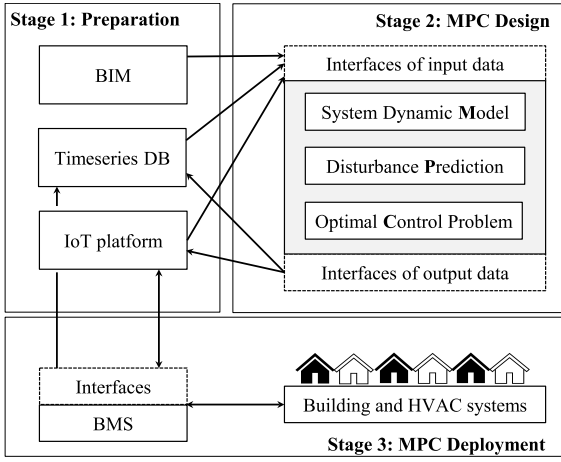


Fig. 1. Typical stages in a real-life MPC application.

suitable modelling approach to describe the system dynamic [3,17], predict the disturbances, and solve the OCP accordingly, dependent on the provided inputs and the required outputs. In this context, the developed MPC algorithm often has complex interfaces tailored for the concrete systems, which is, therefore, difficult to be directly transferred to another building. Eventually, in Stage 3, the validated MPC algorithm will be deployed in the real building operation, based on the existing interfaces of the Building Management System (BMS) or Building Automation System (BAS).

Although all three stages are essential, previous studies have predominantly investigated the second stage in the simulation setting, because the control-oriented model is a knowledge-intensive modelling process and regarded as cornerstone of MPC [3,18]. Nevertheless, a lot of time is often devoted to the other two stages in real practice. Blum et al. [9] conclude in their field MPC study that data preparation and controller deployment consume considerable effort, each accounting for one third, and deserve further investigation. In particular, (1) interpretation and selection of the heterogeneous input data demands expert knowledge and therefore considerable custom engineering [3,5,9]; (2) interfacing of the MPC algorithm demands considerable manual efforts [19,20], which facilitates the MPC parameterization with proper inputs and the communication to the existing BMS with proper outputs, yet has low re-usability among buildings. In other words, one may say that the challenge of practical MPC implementation is, to a large extent, a matter of ineffective data exchange across disparate systems (data silos). Thus, components and processes are crucial for scalable MPC, which can integrate diverse sources, provide a machine-interpretable context of building systems, and allow automated data preparation.

1.2. Semantic data modelling

Semantic data modelling can be defined as creating an explicit representation of data in a way that is accurate and commonly understood by both humans and computers [21]. Among possible representation methods, the ontology or metadata schema refers to the formal conceptualization of classes and relationships of entities in a specific domain [22]. To allow efficient access to such structured data in different systems across different domains, Semantic Web technologies were developed [23], and are used to govern the diverse data consistently. Especially in the Architecture, Engineering and Construction and Facility Management (AEC/FM) industry, the research community has acknowledged Semantic Web technologies as an effective approach to integrate data from disparate domains over the Web [24–27]. Using semantic Web technologies and well-defined ontologies, data of the design phase can be merged with historical or live measurements in the

operation phase for different purposes, such as energy monitoring [28–32], and knowledge-based fault detection and diagnostics in buildings [33–35], and further energy performance optimizations [19,36,37]. Nevertheless, few research studies have systematically discussed what metadata is necessary to support the MPC application, so that we can (semi-)automate the MPC parameterization and reuse the design of interfaces. Also, despite the many ontologies available, how to model the required information using existing open-source data models is still reported as unclear [25,27].

1.3. Outline

This work investigates to what extent Semantic Web technologies can be used to solve the data silo problem confronted in the MPC design. The contributions of this work are three-fold:

1. Design the data model that represents the HVAC system and the MPC algorithm, mainly using openly available ontologies.
2. Create a semantic-graph generator that can populate the proposed data model using real building data (BIM).
3. Demonstrate how the semantic graph can be used to automate the data preparation required by MPC parameterization and deployment.

The rest of the paper is structured as follows. Section 2 describes the standard MPC control scheme and the data sources required, reviews the state-of-the-art ontologies for smart building operation, and semantic graph generators. Section 3 illustrates the proposed terminology model for supporting the MPC application in HVAC systems. Section 4 elaborates on the proposed RDF-graph generator and the prerequisites of the BIM model. Then, Section 5 introduces the use case, which we use to evaluate the proposed data model and graph generator, and explains how the semantic graph can support scalable MPC design. Section 6 discusses the results of the use case. Finally, Section 7 summarizes the main conclusions of this paper and discusses future work.

2. Background

2.1. MPC scheme for building control

The typical MPC formulation for building control can be expressed as an optimal control problem (OCP) [38] by Eq. (1), simplified based on [5].

$$\min_{x,u,y,a} \sum_{k=0}^{N-1} l_k(x_k, u_k, y_k, a_k, r_k, d_k) \quad (1a)$$

$$s.t. \quad x_0 = \hat{x}(t) \quad (1b)$$

$$x_{k+1} = f_{zone}(x_k, u_k, d_k) \quad (1c)$$

$$y_k = g(x_k, u_k, d_k) \quad (1d)$$

$$u_k = f_{hvac}(x_k, a_k, m_k) \quad (1e)$$

$$d_k = d(t + kT_s) \quad (1f)$$

$$r_k = r(t + kT_s) \quad (1g)$$

$$x_k \in \mathcal{X}, u_k \in \mathcal{U}, a_k \in \mathcal{A}, k \in \mathbb{N}_{[0,N-1]} \quad (1h)$$

where $x_k \in \mathbb{R}^{n_x}$ denotes the states of the building, $u_k \in \mathbb{R}^{n_u}$ the control inputs of the zone side, $y_k \in \mathbb{R}^{n_y}$ the control outputs, $a_k \in \mathbb{R}^{n_a}$ the action of the field actuators, $r_k \in \mathbb{R}^{n_r}$ the reference signals, $d_k \in \mathbb{R}^{n_d}$ the disturbances, $\hat{x}$ the estimated states, $m_k \in \mathbb{R}^{n_m}$ the states of the HVAC systems, at the k th step into the prediction horizon T discrete as N steps in total, with a control sampling time of T_s , at a given time t .

Eq. (1a) represents the objective function, where $l_k(x_k, u_k, r_k, d_k)$ is addressed as stage cost and serves to assign a cost to the particular choice of u_k . Eq. (1b) initializes the optimization problem at each time step t with estimated states $\hat{x}$. Eq. (1c) is the system dynamic function and predicts future states, while predicted control outputs are determined via Eq. (1d). The control inputs are subjected to the HVAC operating conditions, formulated as a physical constraint by Eq. (1e). Eq. (1f) refers to the disturbance prediction considered in the MPC, such as the ambient weather and the occupants, while the reference signals mean the forecasted reference, such as the predicted allowed temperature bounds described in Eq. (1g). The bounded constraints on the states, control variables, and actuators are formulated by Eq. (1h).

Among real MPC applications, metadata of the design stage is often required. A typical example of the thermal envelope model (Eq. (1c)) is the Resistance Capacitance (RC) model, which requires metadata about the building topology and envelope material information. The HVAC component equation (Eq. (1e)) usually requires some empirical parameters, nominal conditions and topology information about the pipes, fans, heating/cooling coils and so on, which initially come from the design document of HVAC systems or the product catalog. Similarly, metadata about the buildings and HVAC systems are also taken to set up the constraints of control problems (Eq. (1h)). Although some studies have tried to extract such design information from BIM automatically to create the control model, researchers have implemented their own data structure [39] or non-open-source toolchain [40,41], which are difficult to reuse among buildings. Furthermore, metadata involved in the operation phase, especially the sensor data points, is used to update the system states (Eq. (1b)), measurements (Eq. (1d)) and execute the optimal control. In most existing applications, the connection to BMS systems is often tailored according to the specific system architecture, which causes considerable efforts to transfer the algorithm to another room or building even with similar characteristics. Therefore, the standard data model for such metadata can increase the reusability of MPC algorithm and simplify its configuration.

2.2. Semantic data modelling towards scalable control design

Semantic data modelling refers to the process of describing metadata using ontologies so that semantic models are well-structured and can be interpreted by machines without human intervention. Standardized by the World Wide Web Consortium (W3C), a series of semantic modelling recommendations have been published, such as the Resource Description Framework (RDF) [42], Web Ontology Language (OWL) [43], SPARQL Protocol and RDF Query Language (SPARQL) [44], and Shapes Constraint Language (SHACL) [45]. The classes and relations defined in an ontology are called the Terminology model (T-box). In contrast, the semantic data model for the specific cases (i.e., instances) is addressed as the Assertion model (A-box). Using these technologies, metadata from different domains can be aggregated and exchanged over the Web more efficiently. Semantic Web technologies can, therefore, help to integrate diverse data sources and enable more transferable data-driven services, such as MPC.

Some studies have provided valuable insights about simplifying control configuration using semantic data modelling. The identified contributions lie in two aspects. First, metadata can provide the uniform context of data points in BMS and thus help developers to write more reusable scripts [46,47]. Second, studies [35,48,49] show that by formulating the control rules in semantic data model, control algorithms can be instantiated among buildings swiftly. They demonstrate the promise of semantic modelling for more portable control. However, this solution is more suitable for scaling up rule-based control delivery and does not profit much from model-based control design. We believe using the rich semantic data for control-oriented modelling can unleash the potential of real MPC applications. For instance, the RC model structure, as well as the plausible range of the parameters, can be determined based on the building geometry and material data [41,50].

Table 1
Ontologies reviewed in the study.

Category	Ontology
Building	ifcOWL [52]
	Property Set Ontology [53]
	Building Topology Ontology [54]
	Building Element Ontology [55]
HVAC	Distribution Element Ontology [56]
	Flow System Ontology [57]
	Tubes System Ontology [58]
Sensors	Sensor observation sample and actuator, and Semantic Sensor Network Ontology [59]
	Brick Schema [60]
	Hypermedia Controls Ontology [61]
	Things Description Ontology [62]
	IoT-Stream ontology [63]
Control	Procedure Execution ontology [64]
	CTRLont [65]

Furthermore, some physical values are often required in the system models yet missing in the actual measurements. For instance, the thermal power delivered by individual radiators is required by both performing the system identification for the zone thermal envelope and establishing a radiator model. But only a single thermal meter of the entire hydraulic heating system is installed in most existing buildings [51]. A well-informed estimation of thermal power requires both measurements and system topology. To this end, a standard data model, which can describe the building geometry data, HVAC system topology, and sensor measurements, is crucial for developing reusable MPC design pipelines and interfaces.

2.3. State of the art ontologies

Systematic reviews of ontologies in the AEC/FM domains can be referred to [25,27]. Despite the many ontologies available, Pritoni et al. [25] conclude that building modellers can still find it unclear to select existing suitable ontologies to describe the energy optimization use cases. Building a data model that aligns with the well-defined existing ontologies and can describe the metadata dealt with in the MPC design is worth further investigation. We review the openly available ontologies in building, HVAC systems, sensors and actuators, and control, as listed in Table 1

In terms of building and its product, ifcOWL, Property Set Ontology (PROPS), Building Topology Ontology (BOT), and Building Element Ontology (BEO) are reviewed. IfcOWL [66] is an OWL representation of the most widely-used open BIM standard Industry Foundation Classes [67] (IFC). ifcOWL is more often used to represent building architecture objects and construction data. PROPS translates the property sets PSet defined in the IFC schema into datatype properties in RDF so that the attributes of a BIM object can be easily transferred using the ontology. BOT [68] provides minimum generic concepts for depicting the topological relationships of sub-components in buildings, such as the story, zones, and building elements. BOT serves as the core ontology, while additional vocabularies often enrich detailed information about the geometry and components. For example, BEO [55] is the RDF translation of the class `IfcBuildingElement` from defined in the IFC standard, which offers type information for the building components, e.g., walls, and windows.

In terms of HVAC systems, we review the distribution element ontology, Flow System Ontology (FSO), and TUBES System Ontology (TSO). Distribution Element Ontology is based on the `IfcDistributionElement` subtree in the IFC, and describes a taxonomy definition of common distribution elements. Because such components are often related to the Mechanical, Electrical, and Plumbing (MEP) systems, Distribution Element Ontology is abbreviated as MEP ontology. FSO represents classes and relationships describing the thermal and mass

transfer in a flow system and the composition of such a system [69]. Like FSO, TSO can also represent the composition of pipe and duct systems in buildings. Yet, FSO focuses more on the hierarchical and functional perspectives of the systems that provide interconnected building services.

Regarding sensors, actuators and their data points in the BMS, (1) Semantic Sensor Network (SSN) [70] and the Sensor, Observation, Sample, and Actuator (SOSA) (2) Brick schema (Brick) (3) Web Of Things Things Description Ontology (TD) (4) Web Of Things Hypermedia Controls Ontology (HCTL) and (5) IoT-Stream ontology are reviewed. The SSN [70] and its light-weighted ontology SOSA [71] represent the sensors, actuators, their measurements, and the link to the observed entities coherently. Due to its concise and generic design, SOSA/SSN is recommended by W3C and widely used for sensor network systems in different domains. Brick provides a portable vocabulary for describing building equipment entities, such as HVAC systems, lighting, and fire safety equipment. On the other side, it provides a hierarchy for data points of sensors, actuators, and controllers (e.g., set points, commands, and alarms) in the building domain. Furthermore, Brick provides a reference link to external resources such as time series objects, which recorded the historical measurements. As Internet of Things (IoT) devices are emerging in smart buildings, ontologies such as IoT-Stream [72] and TD [62] are published to define the vocabulary for IoT platforms, devices, services, and stream data. IoT-Stream annotates raw and aggregated data, describing the metadata for search and real-time processing. TD ontology provides a standardized way to model the interactions of IoT devices and services, which is the contextual building blocks of the Web of Things (WoT) Architecture [73] by the W3C Web of Things Working Group. WoT has an ontology specific for reifying the media sources over the internet, namely the HCTL. HCTL describes the links and forms used in RESTful Web applications in a structured way to provide more detailed resource context about sensors, actuators, and the software artifacts used in the IoT environment.

In terms of control in the building systems, the Semantic Energy Aware Systems ontology (SEAS), especially its sub-modules Procedure Execution ontology (PEP), and the CTRLont are reviewed. SEAS [74] is a modular ontology offering high-level vocabulary for smart energy applications. Its sub-module, the PEP ontology [64], generalizes the vocabulary involved in a process, such as Input, Out, and the ProcedureExecutors. Two other SEAS sub-modules, Optimization [75] and Forecasting [76] ontologies refine the concepts in optimization and forecasting processes, which provide the basic vocabularies for describing the MPC procedure. CTRLont [65] is an ontology that formally represents control logic in Building Automation Systems (BAS), to enable better integration, reuse, and automated verification of control logics. While CTRLont is proven effective for rule-based control logic design, it is not suited for model-based optimization, such as MPC, due to its different underlying approach.

Across all the domains above, the Quantities, Units, Dimensions, and Data Types (QUDT) ontology [77] is used to describe the physical quantities and units of measurements to refine the measurements.

2.4. RDF generators for buildings and systems

Well-defined semantic graphs of real buildings are crucial for enabling more transferable smart control applications. However, manually creating such semantic graphs for buildings and HVAC systems in real projects requires considerable time and resources.

BIM is a comprehensive data source for the building design stage and, therefore, used for semantic graph generation. Among different BIM formats, the Industry Foundation Classes (IFC) is the most widely used open BIM specification, and three IFC-based RDF graph generators are available. IFCtoRDF [78] focuses on translating the IFC data exhaustively to RDF graphs, which contain complete IFC data yet are too complex to be directly used in building energy use cases. The improved

converter, IFCtoLBD [79], transforms only the necessary information into RDF graphs, using ontologies recommended by the Linked Building Data (LBD) Community Group. Since the IFCtoLBD converter focuses on the building structure, products, and properties, the converter does not parse HVAC systems and components in the latest release version (2.43.5 Beta). Adding such a feature would be useful, considering the importance of HVAC systems and components in building data modelling and beyond. IFC2TSO [80] can process the IFC file and translate both HVAC systems and simple building topology information into RDF graphs. The successful functioning of the tool relies on a correctly exported IFC file, which must contain a detailed classification of HVAC components. Nevertheless, because some functional concepts required in the HVAC system modelling are missing in the current IFC schema, additional data structured in a JSON file needs to be provided manually, making the preparations for the graph generation work-intensive. To avoid the modelling limits subjected to the IFC standard itself, a further semantic graph generator is developed for the specific BIM-authoring tool. RevitToRDFConverter [81] is a plugin that transforms proprietary Revit data into an RDF graph. The conversion focuses on the HVAC components, as well as the interaction between HVAC components and spaces. However, the generated graph represents mainly design data, and neither sensor nor actuator information is included. About the modelling the data points in BMS systems, Brick developers offer software tool to map the entities defined in tabular format such as a CSV spreadsheet into an RDF graph [82]. The raw data needs to meet some specific column structure, and thus, it takes some manual effort to prepare the input data, depending on the individual BMS. Brick also offers a Rule-Based Model Builder [83] to infer data point type based on BMS labels, which is difficult to generalize due to the diverse naming conventions in BMS.

To summarize, the existing RDF-generators establish a good foundation for transforming data from the design stage into semantic graphs; however, the generated graphs do not meet the requirements for MPC configuration yet. To narrow the gap, we develop an RDF-parser based on the existing RevitToRDFConverter, which includes the information about building geometry as well as sensor and actuators required in control design. We decide on the Revit-based workflow, because Revit provides the native data environment in the design and engineering process, and allows to fix the modelling errors directly to ensure the graph quality.

3. Information model design

We design the T-box using openly available ontologies and make only minimum extensions, to provide a reusable data model for MPC applications in buildings. The conceptualization deals with four main domains and their interactions: control procedures, HVAC systems, buildings, and sensors and actuators. The overview of the proposed T-box is illustrated in Table 2, with the types of information and the corresponding ontologies listed. A basic introduction to the deployed ontologies can be referred to Section 2.3. Four extension ontologies are made to align with the selected open-source ontologies, as marked with 'a'. SSN-EXT extends from SSN to describe the static properties, which do not change along the building operation. BOT-EXT extends BOT to introduce further topological classes or relations. FSO-EXT extends FSO to provide some type information. CONTROL extends PEP, the sub-module of SEAS, to refine the MPC procedure.

The rest of this section provides a detailed elaboration on the proposed data model. Section 3.1 presents the competency questions defining the scope of the information model. Sections 3.2 to 3.5 explain how to individually model the required domain information. A few notes for the graphical illustration are worth mentioning. We depict the entities of MPC algorithm, HVAC systems, buildings, sensors and actuators in blue, light orange, orange and yellow, respectively. The solid rounded rectangle denotes class or its instance, the dashed rectangle for literals, solid arrows for object property, and dashed arrows for datatype property. We use *italic* to highlight the extended vocabularies, and the lowercase of the ontology abbreviation as the corresponding namespace's prefix.

Table 2
Types of information found per domain, and corresponding ontologies.

Domain	Type of information	Deployed Ontology
Control algorithms	Control procedures	SEAS, CONTROL ^a
	Resources	HCTL
	Property	SSN-EXT ^a , QUDT
HVAC systems	Components and hierarchy	FSO, FSO-EXT ^a , MEP
	Topology	FSO
	Functionality	TUBES
	Interaction with the building	FSO
	Property	SOSA/SSN, SSN-EXT ^a , QUDT
Buildings	Element and hierarchy	BOT, BEO, BOT-EXT ^a
	Topology	BOT
	Property	PROPS, SSN-EXT ^a , QUDT
Sensors and actuators	Type and hierarchy	Brick
	Measurements	Brick, TD, HCTL, SOSA/SSN
	Property	SOSA/SSN, QUDT

^a Extension ontologies.

3.1. Competency questions

The following competency questions (CQ) list the necessary information for MPC implementation, which covers the data involved in the data points' interpretation, control-oriented modelling, and closed-loop control processes. The final information model is expected to answer all CQs to be considered functional and complete.

1. Which controller implements the MPC algorithm and what exact procedures does it deploy?
2. What properties do the aforementioned procedures have and how are these procedures connected to the existing IT infrastructure?
3. What are the required inputs and outputs of the controller for the MPC algorithm?
4. Which HVAC device and building does the MPC algorithm serve?
5. Which HVAC component and system information is available and needed for the design of the MPC algorithm?
6. Which functionality do the HVAC systems support within buildings?
7. What is the topological, building-physical, and geometric information for the spaces, systems and elements in the building?
8. Which sensors and actuators are associated with which devices and building zones?
9. How are the sensors and actuators connected to virtual data points, and where are the data points storing their data?

To answer the CQs thoroughly, the information is classified into four parts: MPC algorithm (CQ1-4; Section 3.2), HVAC systems (CQ5-6; Section 3.3), building structure and geometry (CQ7; Section 3.4), as well as sensors and actuators (CQ8-9; Section 3.5).

3.2. MPC algorithm

In general, a MPC algorithm is realized by a specific controller and consists of three independent procedures, namely performing system identification, making forecasts, and solving an optimal control problem (CQ1). We suggest these procedures to be supported by the ontologies displayed in Fig. 2. The extension `control:` is proposed to represent the three mentioned MPC procedures. Design properties of these procedures are described under the namespace `ssn-ext:`. Each procedure is further linked with the corresponding Uniform Resource Locators (URLs) via HCTL, to be able to support the algorithm deployment potentially as a web service. Last, the TD ontology is used to model the IoT-related information. Throughout the below section, references are made to the MPC variables that were explained in Section 2.1.

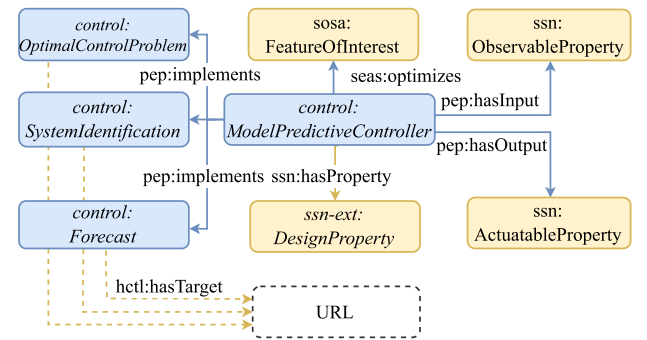


Fig. 2. T-box for the MPC algorithm.

3.2.1. The MPC controller

`control:ModelPredictiveController`, defined as the subclass of the `seas:OptimizationExecutor`, represents the controller that executes a specific MPC algorithm (CQ1). It connects to the entity `sosa:FeatureOfInterest` that the algorithm serves via `seas:optimizes` (CQ4). The abstract `sosa:FeatureOfInterest` class allows for the modelling of different use cases, such as instance of a thermal zone `bot:Zone` or a specific component `fso:Component` like the chiller. The controller is linked to the measurable properties of room side y_k or HVAC side m_k via `pep:hasInput`, and to the actuable properties of the actuators a_k via `pep:hasOutput` (CQ3). y_k are expected to be measurable `ssn:ObservableProperty` instances, which are observed by the corresponding sensors (`sosa:Sensor`). The a_k are controllable `ssn:ActuableProperty` instances, which are actuated by the actuators. The prediction horizon T and the sampling time interval T_s are the two most common design properties of an MPC controller (CQ2), and will be inherited by all child procedures implemented by this controller.

3.2.2. Forecast procedure

Forecasts on the disturbance d (Eq. (1f)) and reference signals r (Eq. (1g)) provide the insights of the future for the controller to finding the optimal control. The disturbances in the building control domain include mainly the weather conditions and occupants. In the context of economic MPC, disturbance can also refer to the price signal of the electrical grid, which is frequently dealt with in the objective function of an OCP for stage cost calculation. The reference signals represent the allowable range of the states, such as the temperature, humidity and CO₂ concentration, subjected to the thermal comfort. Defined as a subclass of `seas:Forecasting`, `control:Forecast` is usually implemented as a set of stand-alone functions for the individual

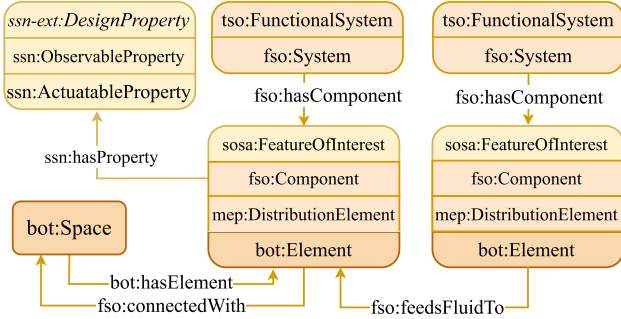


Fig. 3. T-box for the HVAC devices and systems.

forecasting task. Each `control:Forecast` should be linked to the endpoint via `hctl:hasTarget` (CQ2).

3.2.3. System identification procedure

The class `control:SystemIdentification` represents a collection of algorithms for determining the system dynamic model f_{zone} as in Eq. (1c) used in the MPC controller. Furthermore, `control:SystemIdentification` should have two further design properties: training data length and adaptivity (CQ2). As mentioned in [9], the training data length is critical for training a valid dynamic model. `ssn-ext:Adaptivity` with value `True` indicates that the model estimation is an online learning algorithm that updates the model depending on a trigger. Otherwise, a pre-trained system dynamic model will be directly called by the controller, and the system identification procedure will be omitted. The individual SI algorithm is frequently implemented as an independent function, and should ideally be represented by its own endpoint (CQ2).

3.2.4. Optimal control procedure

Defined as the subclass of `seas:Optimization`, `control:OptimalControlProblem` refers to the procedure of solving the OCP formulated as Eq. (1). The procedure involves the initialization of the problem with the corresponding constraints and latest measurements, and then solving the optimization problem with the system dynamic model and forecasts recursively. Therefore, the optimal control procedure should also have access to the results of the SI procedure and forecasts procedure, which are executed by the same controller (CQ1). Each type of OCP targeting different objectives, such as minimizing the energy consumption or maximizing the thermal comfort, should be modelled as an individual instance and linked to the corresponding function via `hctl:hasTarget` (CQ2).

3.3. HVAC systems

The essential metadata of HVAC systems include the properties of the HVAC component itself, and its interaction with the building. To model such comprehensive information, the HVAC entity is usually defined as an instance of different classes from multiple ontologies simultaneously. The proposed model is illustrated in Fig. 3, and explained from component and topology aspects.

On the component level, a HVAC component is firstly defined as an instance of the class `fso:Component`. It can be linked to other components and the served thermal zones via `fso:connectedWith`. Furthermore, subproperties of `fso:connectedWith` such as `fso:exchangesFluidWith` and `fso:exchangesHeatWith` can be used according to the specific thermal/mass flow exchange and flow direction. More detailed usage of FSO can be referred to [69]. Second, the same component is modelled as subclass instances of both `fso:Component` and `mep:DistributionElement`, which provides the taxonomy. For instance, a radiator entity is modelled as

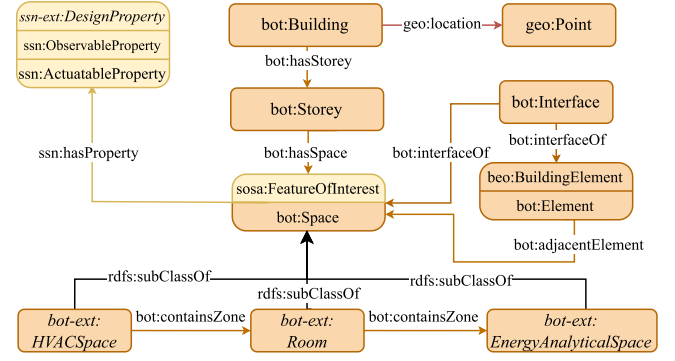


Fig. 4. T-box for the topological information of the building.

a instance of both `fso-ext:Radiator`, an extended subclass of `fso:Component`, and `mep:SpaceHeater`, the subclass of `mep:DistributionElement`. Third, the HVAC entity is also defined as a `sosa:FeatureOfInterest` instance, to model further properties via `ssn:Property`. Static properties, represented by `ssn-ext:DesignProperty`, are usually stable during the complete life cycle, and are specified in the design stage, such as the nominal heating power of the radiator. Dynamic properties vary over time, which can be measured by sensors or actuate by actuators, namely `ssn:ObservableProperty` and `ssn:ActuatableProperty`. Dynamic properties are linked to measurements and sensors/actuators following the SOSA/SSN ontologies. Fourth, the HVAC entity is further declared as a `bot:Element` instance, so that it can be linked to its location `bot:Space` via `bot:hasElement`.

On the system level, we model the system topology using FSO, and functionality using TSO, via defining the HVAC system entity as an instance of both classes `fso:System` and `tso:FunctionalSystem`. By modelling the system (`fso:System`) composition via `fso:hasComponent` instances and flow direction between components with `fso:exchangesFluidWith`, the topology of a system can be retrieved via the available property paths. The sub-classes of `fso:System`, `fso:ReturnSystem` and `fso:SupplySystem`, allow to distinguish components from different sides easily (CQ6). In this way, we can locate the connected components in the upper or lower stream of the same system as well as the sensors attached (CQ5). The sub-classes of `tso:FunctionalSystem`, such as `tso:CoolingSystem` and `tso:HeatingSystem`, provides the functional context of the HVAC system.

3.4. Overall building information

As illustrated in Fig. 4, the BOT ontology is used to describe the topology of the building structure, i.e., a building contains storeys, and storeys contain spaces, represented via `bot:hasSpace` (CQ7). The geo location is linked to the building via `geo:location` using GEO (CQ7). A `bot:Space` is extended with three custom sub-classes: `bot-ext:Room`, `bot-ext:HVACSpace` and `bot-ext:EnergyAnalyticalSpace`, to provide classification of different disciplines. `bot-ext:Room` represents the rooms designed by architects, `bot-ext:HVACSpace` refers to the HVAC functional zones designed by the mechanical engineer, while the `bot-ext:EnergyAnalyticalSpace` serves as the space unit for heat transfer analysis purposes. The connections between these zones can be modelled using `bot:containsZone` relations (CQ7).

In our approach, we model instances of `bot:Space` at the same time as instances of `sosa:FeatureOfInterest`, so that its static and dynamic properties (CQ8) can be further refined, same as in Section 3.3. The static properties of a space include the design heating

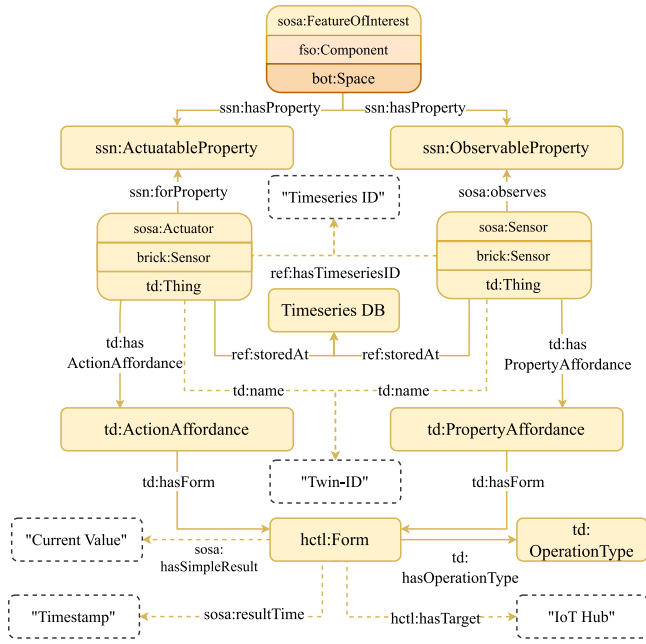


Fig. 5. T-box for the sensors and actuators.

load, cooling load, minimal airflow rate, area and volume, modelled as datatype properties using PROPS (CQ7).

The space boundary is critical for thermal zone modelling, since it determines the heat transfer among zones. These space boundaries can be encoded using `bot:Element` and `bot:Interface` instances. The space is associated to its bounding elements via `bot:adjacentElement`. The instance of `bot:Element` is further refined as a subclass instance of `beo:BuildingElement`, to model the element type information. For instance, a wall entity is an instance of both `bot:Element` and `beo:Wall`. Due to the intersection of bounding elements, each element can be divided into smaller parts as `bot:Interface` for thermal analysis, often addressed as 2nd level space boundaries. Each `bot:Interface` instance is linked to the belonging element as well as each adjacent space via `bot:interfaceOf` (CQ7). Two spaces that share the same bounding elements and opposite interfaces are regarded as adjacent. For thermal analysis, the critical static properties of an interface include the thermal transmittance or heat transfer coefficient U and area-specific heat capacity c , which are modelled using PROPS (CQ7).

3.5. Sensors and actuators

Fig. 5 shows the overall information model used for modelling sensors and actuators. This part relies heavily on the SSN/SOSA ontologies, BRICK ontology, and the TD ontology. The sensors measure the observable states of the systems, while the actuators measure and set the actuation as desired (CQ8). The relationship between the observed `sosa:FeatureOfInterest` instance and its properties is modelled using `ssn:hasProperty`, which stands either for measurable states `ssn:ObservableProperty` such as the room temperature, or for controllable variables `ssn:ActuatableProperty` such as the opening of the valve. The `sosa:FeatureOfInterest` in the context of building HVAC control can refer to the `bot:Zone` and a specific device `fso:Component`, whose data models are explained in the previous sections.

A `sosa:Sensor` instance can be connected to its states via `sosa:observes`, while the actuator `sosa:Actuator` is linked

to the controlled variables using `ssn:forProperty`. To add further quantity information to the measurements, the sensor and actuators are defined also as instances of the corresponding sub-classes of the `brick:Sensor` class (CQ 8), such as a `brick:Temperature_Sensor`. The type of the sensor can later be used to determine the appropriate pre-processing method to be used in the MPC algorithm.

The data flow from the sensors and actuators required during the MPC loop can be categorized into two types: time series data for historical measurements, and streaming time series data for real-time communication (CQ9). The historical data is usually stored in a time series database and used for system identification. The sensors and actuators are connected to the time series database instance via `ref:storedAt` and linked with its unique key via `ref:hasTimeseriesId` to gain access to the data (CQ9). The IoT stream data is expected to come from an IoT hub, routed via the IoT gateway and transferred in the form of IoT messages. Stream data is mandatory for controlling the actuators in run-time. To model such IoT-related information, the TD ontology is selected. The sensors and actuators are modelled as instances of `td:Thing` and have their interactions `td:InteractionAffordance` with the IoT platform, either as `td:PropertyAffordance` of measurement y_k or `td:ActionAffordance` of actuation a_k , respectively. The unique device ID on the IoT platform is modelled as `td:name`. The details about `td:InteractionAffordance` can be refined via `hctl:Form`, which provides the context of the IoT message received (CQ9).

4. Population of the information model

Using the information model as presented in Section 3, it is possible to build RDF instance graphs accordingly. Ideally, a software solution is required to generate an RDF graph automatically for real buildings. Nowadays, BIM models are the most comprehensive information sources for the building design and engineering stage. Nevertheless, real BIM projects often contain many modelling inadequacies and may not meet up the requirements for generating a complete valid semantic graph. We develop a BIM-based RDF graph generator, which allows the developers to fix the modelling errors interactively during the graph generation process. We consider Revit, one of the most popular BIM-authoring tools, in the implementation and take the existing open-source project `RevitToRDFConverter` [84] as the starting point in our development.

4.1. Prerequisite of the BIM project

In order to generate a complete and valid graph, the BIM project must include information of these three categorizations: (1) Well-defined space boundaries and material properties of the building envelope; (2) the classification and typical design properties of the HVAC systems and their components; (3) the classification of sensors and actuators as well as their unique identifier (ID) in the existing infrastructure, such as the time series database and IoT platform.

How to model such information in the specific BIM-authoring tool differs and no standard way exists. In our case of Revit, the following approaches are taken. We utilize the “Energy Optimization” feature to create a detailed thermal envelope model [85], which is qualified for thermal analysis. To provide classification information, we add `IfcExportAs` and `IfcExportType` to all MEP families as “Family Parameter”, and fill in both attributes according to the class definition in IFC standard [86]. For typical HVAC design properties, we model the attributes to the individual MEP family.

4.2. Workflow

Fig. 6 outlines the overall workflow of the proposed RDF graph generator. The graph generator firstly traverses through building elements

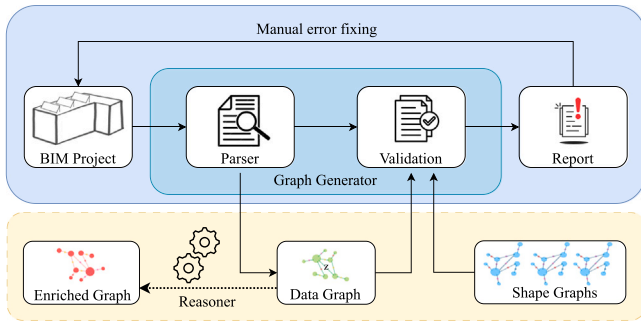


Fig. 6. Graph generation process.

and HVAC components, extracts the required information, and populates the proposed T-box into an instance graph using Turtle [87] as a serialization format. Then, the integrity of the intermediate RDF graph is validated against shape graphs defined using SHACL. A violation report will be sent to the modeller displaying a list of components with modelling errors, if any modelling error exists. After the BIM modeller resolves all modelling errors manually, a valid RDF graph is sent to the graph database. Then, the graph can be enriched with additional information via reasoning, according to the ontologies used and the related alignment ontologies.

We use ontotext GraphDB [88] to manage the RDF data. The violation report is visualized as an interactive window in Revit, which allows the modeller to click on the individual error message and identify the wrongly modelled component. In this way, the modeller fixes the modelling inadequacy iteratively.

4.3. Graph validation

To validate the intermediate graph during the graph generation process, SHACL shapes targeting the HVAC components and sensors classes are defined. They specify the mandatory properties and topological connection of the component instances. Instead of defining generic shapes for the categorization of components, e.g., for all the hydraulic components as in [81], we define a node shape for each component class using property and SPARQL-based constraints. Depending on the user requirements, this allows a more detailed and flexible validation for each component type.

Taking the actuator for example, two rules beneath are defined for the class `sosa:Actuator`. The implementation of the rules is shown in Listing 1.

1. Each actuator has exactly one time series ID;
2. Each actuator has exactly one IoT ID.

```

1 sosa:Actuator
2   rdf:type sh:NodeShape ;
3   sh:sparql [
4     sh:select """
5 PREFIX sosa: <http://www.w3.org/ns/sosa/>
6 PREFIX bric: <https://brickschema.org/schema/
7 -Bric#>
8 PREFIX ref: <https://brickschema.org/schema/
9 -Bric/ref#>
10 PREFIX td: <https://www.w3.org/2019/wot/td#>
11 select $this
12 where {
13   $this a sosa:Actuator;
14     a td:Thing;
15     ref:hasTimeseriesID ?timeseries_ID;
16     td:name ?iot_ID.}

```

```

17 GROUP BY $this
18 HAVING (COUNT(?timeseries_ID) != 1 ||
19 COUNT(?iot_ID) = !1)""";
20 ] ;
21 sh:targetClass sosa:Actuator.

```

Listing 1: SHACL shape for actuator.

In total, 31 SHACL shapes have been developed, including 14 for the heating components, 15 for the ventilation components, and 2 for sensors and actuators. The shapes are defined in three separate shape files correspondingly, which allows the modeller to validate the graph depending on the disciplines in the error-fixing process.

4.4. Alignment ontologies

After successfully validating the RDF graph against the shape graphs, the RDF graph is sent to the graph database. Alignments between the extension ontologies made in this study (`bot-ext`, `control`, `ssn-ext`, `fso-ext`) and the already available ontologies (MEP and BEO) are made and uploaded into the same repository so that the reasoner provided by the database can enrich the graph with inferences.

5. Example graph and use cases demonstration

In this section, example graphs and use cases are presented to showcase the developed RDF graph generator, the data model, and its usage for control configuration. The Turtle, SHACL, SPARQL files mentioned in this study and the RDF graph generator's source code are available on GitHub (details in Appendix A). First, Section 5.1 describes the studied building and its overall graph generation results. Then, Section 5.2 introduces the instance graph of one meeting room to provide the reader with a detailed explanation of the proposed data model. Section 5.3 demonstrates use cases related to MPC configuration using the graph data.

5.1. Graph generation of the building

We take the BIM model of an existing office building and generate its RDF graph using the developed software. The building is located at the Bosch Research Campus in Renningen, Germany. We focus on the three-story office area within the building, which has a total area of 2982 m² and is equipped with a radiator system and a variable air volume (VAV) system. The BIM objects of the office building were successfully translated into 325,519 RDF triples. The complete RDF graph contains data of 80 HVAC zones.

Multiple modelling inadequacies and errors were spotted in the original BIM model. They were fixed based on the validation process and messages provided by the graph generator, as illustrated in Fig. 7. The most frequent modelling errors of MEP systems in BIM include: (1) Improper connection of the components, such as open ports or wrong flow orientation; (2) Duplicated components at the same spatial coordinates; (3) Missing parameters of the components, such as type and nominal conditions.

5.2. Instance graphs of a meeting room

Because the complete graph is too large to demonstrate with details, we present in this section the instance graph (A-box) of one particular meeting room, A.162, which is shown in Fig. 8. The instance graphs related to the building, HVAC systems, sensors, and actuators are automatically generated using the developed graph generator. The instance graph related to the MPC controller is manually created, as such information is not available in the BIM model but is provided by the control engineer separately.



Fig. 7. Graph validation results during the graph generation. The line marked in blue shows the error that the element with ID 1729060 (a fire damper) has no IfcExportedAs attribute.

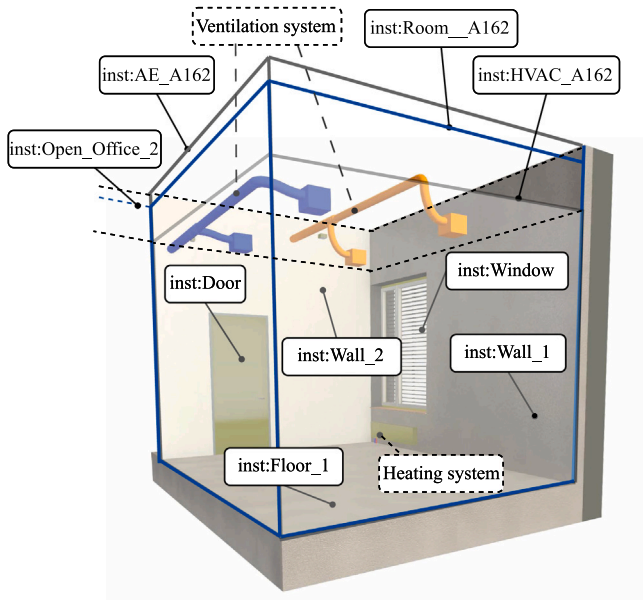


Fig. 8. 3D illustration of the meeting room A.162. It has a radiator and a VAV box with a blue supply branch and a yellow return branch.

5.2.1. Building information

Fig. 9 displays the part of A-box related to the building information. The `bot-ext:HVACSpace` of room A.162 refers to the volume beneath the suspended ceiling, which is the actual air-conditioned volume. It has a volume of 55 m³ and an area of 18.4 m². Important design properties of an HVAC space include supply air volume flow, return air volume flow, heating demand, and cooling demand. Furthermore, the `bot-ext:Room` volume of the room A.162 is bounded by walls (`inst:Wall_1`, `inst:Wall_2`), the floor (`inst:Floor_1`), the door (`inst:Door`), the window (`inst:Window`) and the floor of the room above.

The room `inst:Open_Office_2` is located next to the room `inst:Room_A162`, both of which are adjacent to the same building element `inst:Wall_2`. The related surfaces for heat transfer analysis are refined via `bot:Interface` and linked to the `bot-ext:EnergyAnalyticalSpace` instances. Taking `inst:Wall_2` as an example, on the side facing `inst:AE_A162` it has an interface `inst:AE_Surface_2_1`, while on the other side facing

`inst:AE_Open_Office_2` it has an inverse interface of `inst:AE_Surface_2_2`. The connection between the interface and its counterpart are described via `bot-ext:inverseOf`. We also model the thermal properties of the individual surfaces. For instance, `inst:AE_Surface_2_1` stands for an interior wall with an area of 18.3 m², a tilt of 90°, an orientation of south, C value of 18 000 $\frac{\text{J}}{\text{m}^2 \text{K}}$ and U-value of 1.73 $\frac{\text{W}}{\text{m}^2 \text{K}}$.

5.2.2. HVAC systems and sensors

A detailed view of HVAC components and the connected sensors and actuators is available in Appendix B. Its RDF representation is displayed in Fig. 10. Starting from the HVAC system, the supply air is first treated by `inst:ReheatCoil`, flows through the ducts, and eventually distributed into `inst:HVACSpace_A160` via two air terminals `inst:AirTerminal_1` and `inst:AirTerminal_2`. The supply air volume flow is controlled via `inst:Damper_1` installed in the supply branch. The return branch has a similar topology. The interaction between the air terminals and the HVAC zone is represented by `fso:feedsFluidTo`, while the heat exchange between `inst:Radiator_1` and the zone is described by `fso:transferHeatTo`. Nominal properties of the components are modelled as instances of `ssn-ext:DesignProperty`. Each design property refers to the unit and the actual value using the `brick:value` and `brick:hasUnit`. modelling of `inst:Damper_1` with a diameter of 125 mm is shown as an example.

To control such a room air-conditioned by a VAV box and a radiator, smart sensors, and actuators must be installed on the radiators, reheating coil, dampers, and ducts. Furthermore, both an IoT platform and a time series database are required to track historical and real-time data for closed-loop control. Room A.162 has already installed the required sensors and actuators connected to the InfluxDB and Azure IoT hub. Fig. 10 presents the modelling of two data points, namely the supply air temperature $\vartheta_{z,1}$ of the room and the water damper position $\alpha_{rhc,w}$ of the reheating coil, as an example. Note that a data point may not be directly linked to the object that it describes. For instance, the measurement of $\vartheta_{z,1}$ of room A.162 is connected to `inst:Duct_2`, where the actual sensor is installed. Therefore, concrete topology information of the HVAC systems is crucial to locating the correct data points for further analyses.

5.2.3. MPC controller

As this study targets designing the interfaces reusable for MPC algorithms, we focus on modelling the procedures in the MPC algorithm and the links to the building and HVAC instances involved. A generic use case is assumed for room A.162: the objective of the MPC

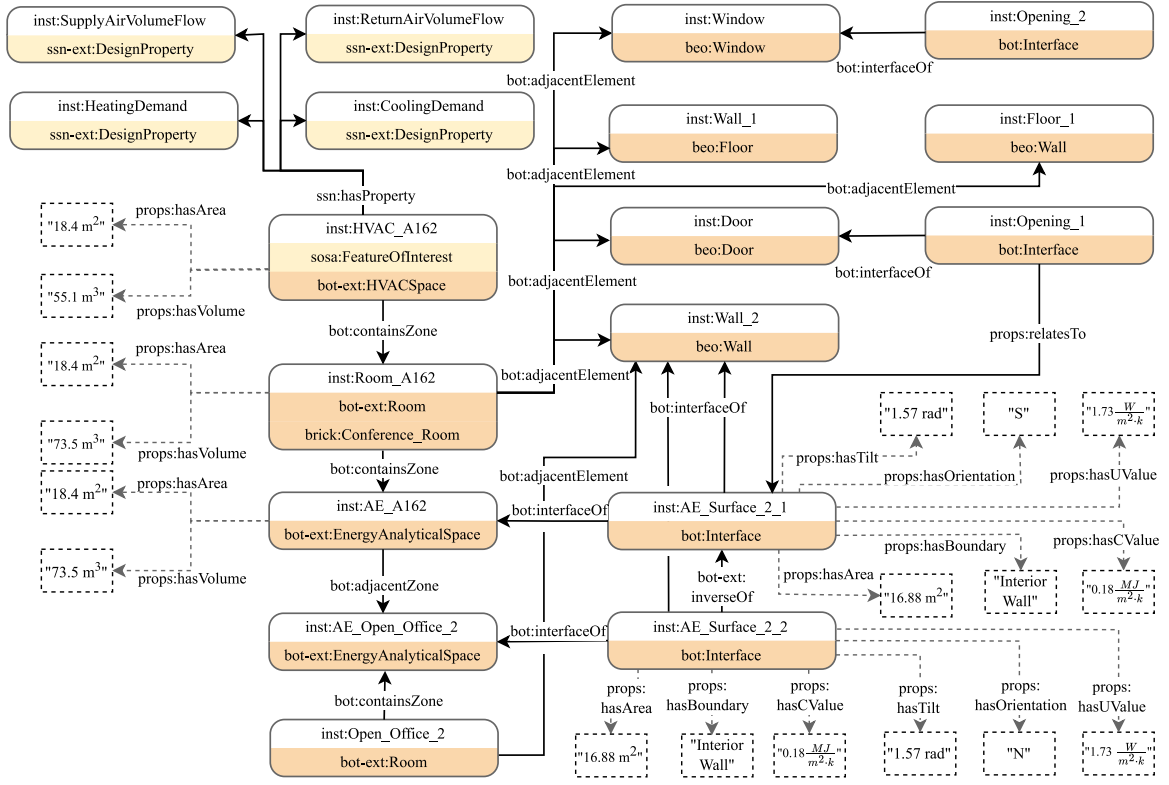


Fig. 9. A-box of the overall building information.

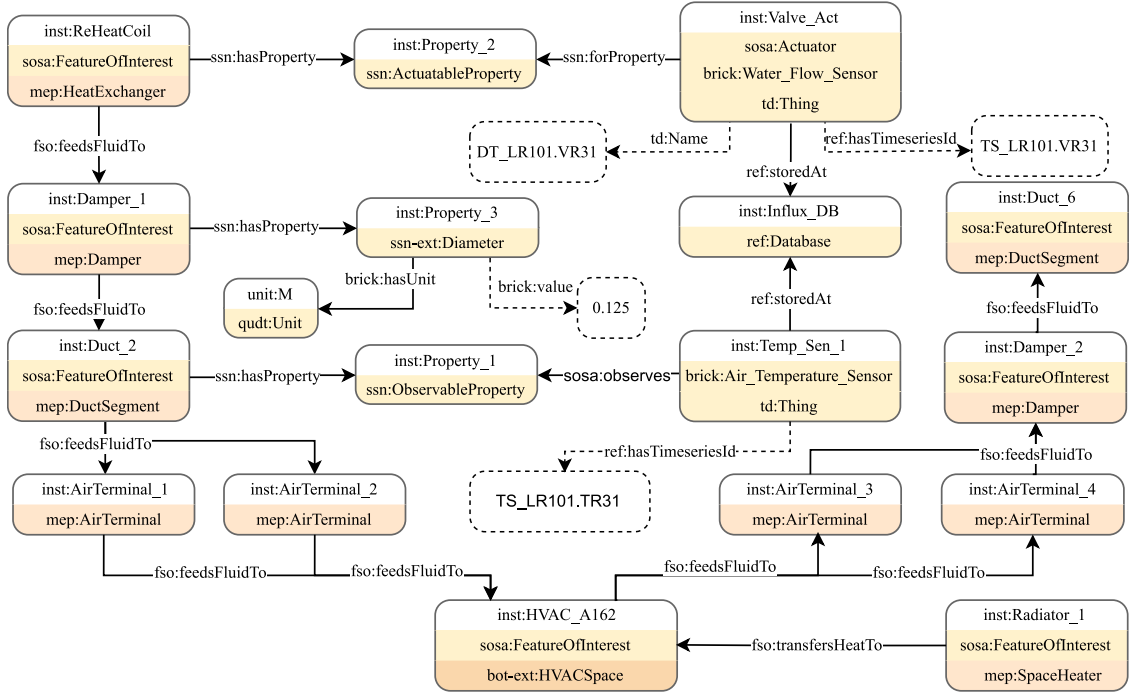


Fig. 10. A-box of the HVAC systems and sensors of room A.162. Only the key components are displayed for simplification purposes, while the connection components, such as fittings, are omitted.

algorithm is to minimize the energy consumption of the room-side HVAC system, and a Resistance and Capacitance (RC) is used as the system dynamic model. The controller optimizes inputs for the VAV box and the radiators. More specifically, MPC algorithm determines the optimal actuation of the following variables: the water valve position

$\alpha_{rh,c,w}$ of the reheating coil in the VAV box, the supply damper position $\alpha_{damp,1}$ and return damper position $\alpha_{damp,2}$ in the VAV box, and the supply water temperature $\theta_{rad,1}$ as well as the water flow rate $\dot{m}_{rad}$ of the radiators, to achieve the control objective. We refer to existing studies for required metadata and the concrete modelling methods,

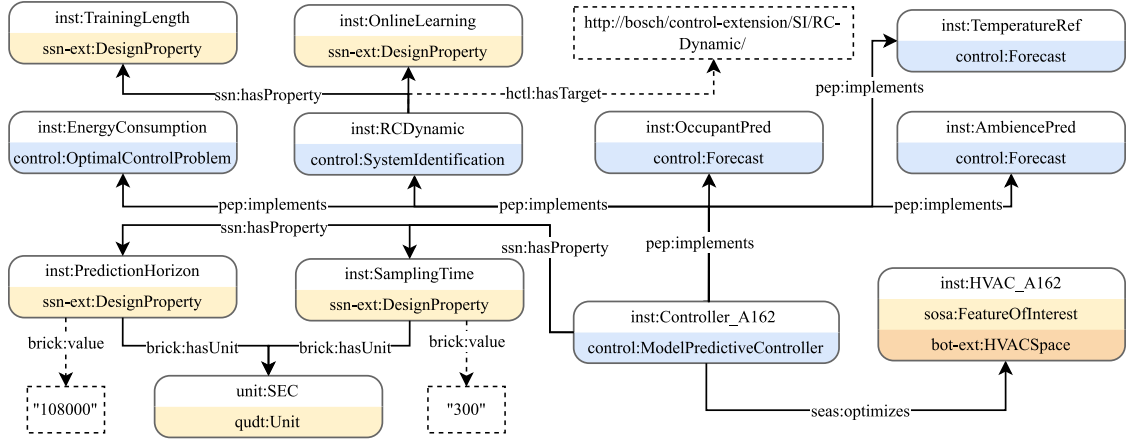


Fig. 11. A-box of the MPC controller of room A.162.

i.e., dynamics for building envelope using RC model as [50], for the reheating coil as [10], and the radiators [89,90]. The implementation of the MPC algorithm is out of scope of the current study.

As shown in Fig. 11, *inst:Controller_A162* is connected to the optimized zone *inst:HVAC_A162* via *seas:optimizes*. *inst:Controller_A162* has two design properties: prediction horizon T and control sampling time T_s , which are 108,000 s (3 h) and 300 s (5 min). The controller implements three types of procedures: system identification, forecasting of disturbances, and optimization, which inherit the design properties of the controller. The system identification procedure *control:RCDynamic* utilizes the RC model and has two further design properties: the training length and the online learning, as suggested in [50]. *control:RCDynamic* is linked to a URL via *hctl:hasTarget*, which can be used as the endpoint for calling the system identification algorithm as web service. As for the forecasts, the occupant disturbance *control:OccupantPred* and weather disturbance *control:AmbiencePred* are considered. As for the optimal control problem, *control:EnergyConservation* refers to the control formulation that minimizes energy usage. Similar to the system identification procedure, each particular optimization and forecast instance has its unique URL as an identifier for the individual service.

5.3. Graph-based data preparation for control

As mentioned in Section 1, MPC algorithms of buildings are difficult to scale due to the heterogeneity of building data and lack of reusability of the interfaces. This section presents the potential use cases for simplifying the MPC configurations based on the RDF graph, in which the important metadata about the building and its HVAC systems is encoded.

5.3.1. Querying geometry of the thermal zone

Information about space boundary, and thermal properties of the surfaces is crucial and often used to set up the dynamic model of the thermal envelope. More specifically, space boundary and geometric information are often required to perform a valid system identification: For the RC model, they are used to determine the model structure [91], the initial value and the boundary of the parameters [50]; for neural network-based models, they influence the network structure [92, 93] and initial parameters [92]. Such information can be queried as displayed in Listing 2.

```
1 SELECT DISTINCT ?element ?eleArea ?AESurface
2 ?surBoundary ?surArea ?uValue ?cValue
3 (SUM(?AEOpeningArea) AS ?totalOpeningArea)
```

```
4 (GROUP_CONCAT(?AEOpening;separator="\n") AS
5 ?beoOpeningList)
6 WHERE
7 {
8 ?Zone a bot-ext:HVACSpace;
9   bot:containsZone+ ?AESpace;
10  props:hasRevitId '1500542'.
11 ?AESpace a bot-ext:EnergyAnalyticalSpace.
12 ?AESurface a bot:Interface;
13   props:hasArea ?surAreaWithUnit;
14   props:hasBoundary ?surBoundary;
15   props:hasUvalue ?uValueWithUnit;
16   props:hasCvalue ?cValueWithUnit;
17   bot:interfaceOf ?element;
18   bot:interfaceOf ?AESpace.
19 ?element a bot:Element;
20   props:hasArea ?eleAreaWithUnit.
21 OPTIONAL
22 {?AEOpening a bot:Interface;
23   props:relatesTo ?AESurface;
24   props:hasHeight ?openingHeightWithUnit;
25   props:hasBoundary ?openingBoundary;
26   props:hasWidth ?openingWidthWithUnit.
27 BIND(xsd:float(STRBEFORE(?openingHeightWithUnit,
28   "")) AS ?openingHeight)
29 BIND(xsd:float(STRBEFORE(?openingWidthWithUnit,
30   "")) AS ?openingWidth)
31 BIND(?openingHeight*?openingWidth
32 AS ?AEOpeningArea)}
33
34 BIND(xsd:float(STRBEFORE(?surAreaWithUnit,
35   "")) AS ?surArea)
36 BIND(xsd:float(STRBEFORE(?uValueWithUnit,
37   "")) AS ?uValue)
38 BIND(xsd:float(STRBEFORE(?cValueWithUnit,
39   "")) AS ?cValue)
40 BIND(xsd:float(STRBEFORE(?eleAreaWithUnit,
41   "")) AS ?eleArea)
42 }
43 GROUP BY ?element ?eleArea ?AESurface ?surArea
44 ?surBoundary ?orientation ?uValue ?cValue
```

Listing 2: Query for thermal and geometric information.

Table 3 shows the results of the targeted room A.162 (with Revit ID 1500542). It displays the properties of the individual thermal surfaces and their belonging building elements. Six building elements are listed under the “Element” column, while eight energy analytical surfaces are listed in the “Interface” column. Namely, *inst:Wall_3* represents an interior wall adjacent to a split level with three corresponding child energy analytical surfaces. Note that the element’s area equals the sum

Table 3

Detailed geometric and thermal properties of surfaces of room A.162.

Element	Element area (m ²)	Interface	Boundary	Interface area (m ²)	U-value (W/m ² K)	c-value (MJ/m ² K)	Opening	Opening area (m ²)
inst:Wall_1	45.3	inst:AE_Surface_1_1	Exterior wall	23.0	0.24	0.36	inst:Window	4.1
inst:Floor_1	992.7	inst:AE_Surface_7	Interior floor	20.7	0.2	0.44	–	0
inst:Wall_2	25.9	inst:AE_Surface_2_1	Interior wall	18.3	1.73	0.18	inst:Door	2.2
inst:Wall_3	234.5	inst:AE_Surface_3_1	Interior wall	6.9	1.73	0.18	–	0
inst:Wall_3	234.5	inst:AE_Surface_4_1	Interior wall	9.1	1.73	0.18	–	0
inst:Wall_3	234.5	inst:AE_Surface_5_1	Interior wall	2.2	1.73	0.18	–	0
inst:Wall_4	19.6	inst:AE_Surface_6_1	Interior wall	23.0	1.73	0.18	–	0
inst:Ceiling_1	992.7	inst:AE_Surface_8	Interior floor	20.7	0.2	0.44	–	0

Table 4

Data points of the radiator.

Data points	Time series ID	Twin ID
$\dot{m}_{rad}$	TS_HW002_1.AR01	DT_HW002_1.AR01
$\theta_{rad,1}$	TS_HW002_1.TR51	DT_HW002_1.TR51

of its related interfaces on each side. As one building element, such as a piece of a wall, can span over multiple rooms, the interfaces involved in one specific space can be a subset of all its interfaces. inst:Wall_1 and inst:Wall_2 have the openings inst:Window and inst:Door as shown in Fig. 8, while the other surfaces are entirely opaque. As the HVAC space has a surface and a window exposed to the outdoor ambience, the configuration of the dynamic model needs to include both the outdoor temperature and solar irradiation as disturbances.

5.3.2. Querying HVAC component and its properties

Crucial type of the HVAC components' metadata for MPC include: (1) the nominal properties, used to parameterize the control model or to set up the constraints in OCP; (2) the detailed topology of the HVAC systems, used to locate the corresponding sensors and actuators connected to components in upstream or downstream; (3) the data points in the existing BMS infrastructure, used to enable automated data exchange. Listing 3 takes the radiator as an example and shows how to query the required nominal and dynamic properties. Consisting of two sub-queries, the first sub-query targets the nominal properties and data points directly linked to the radiator(s). The second sub-query locates the sensor that measures the radiator's supply water temperature. In practice, the supply water temperature is not measured individually at each radiator but only at the main branch of the heating distribution system. Therefore, the heating system topology is used to locate the temperature sensor at the radiator's nearest upper stream.

```

1 SELECT DISTINCT ?radiator ?actuatorID
2 ?tempSensorID ?actuatorIoT ?tempSensorIoT
3 ?nominalHeatingPowerValue ?nominalSupTempValue
4 ?expoCoefficientValue ?nominalRetTempValue
5 WHERE {
6 {SELECT ?radiator ?actuatorID ?actuatorIoT
7 ?nominalHeatingPowerValue ?nominalSupTempValue
8 ?expoCoefficientValue ?nominalRetTempValue
9 WHERE
10 { ?space a bot-ext:HVACSpace;
11 props:hasRevitId '1500542' .
12 ?radiator fso:transfersHeatTo ?space;
13 a fso-ext:Radiator;
14 a sosa:FeatureOfInterest;
15 ssn:hasProperty ?flowProperty;
16 ssn:hasProperty ?tempSetpoint.
17 ?actuator ssn:forProperty ?flowProperty;
18 a brick:Water_Flow_Sensor;
19 td:name ?actuatorIoT;
20 ref:hasTimeseriesID ?actuatorID.
21 ?radiator ssn:hasProperty ?nominalHeatingPo
22 wer, ?nominalSupTemp, ?nominalRetTemp, ?expoCo

```

```

23 -efficient.
24 ?nominalHeatingPower a ssn-ext:NominalPower;
25 brick:value ?nominalHeatingPowerValue.
26 ?nominalSupTemp a ssn-ext:NominalSupplyTempe
27 -rature;
28 brick:value ?nominalSupTempValue .
29 ?nominalRetTemp a ssn-ext:NominalReturnTempe
30 -rature;
31 brick:value ?nominalRetTempValue .
32 ?expoCoefficient a ssn-ext:ExponentialCoeffi
33 -cient;
34 brick:value ?expoCoefficientValue.}
35 }
36 {SELECT (COUNT(DISTINCT ?step) as ?distance)
37 ?tempSensorID ?tempSensorIoT
38 WHERE
39 { ?space a bot-ext:HVACSpace;
40 props:hasRevitId '1500542' .
41 ?radiator fso:transfersHeatTo ?space;
42 a fso-ext:Radiator ;
43 a sosa:FeatureOfInterest .
44 ?component fso:feedsFluidTo* ?step;
45 a fso-ext:SensorFitting;
46 a sosa:FeatureOfInterest.
47 ?step fso:feedsFluidTo+ ?radiator.
48 ?sys a fso:SupplySystem;
49 fso:hasComponent ?component, ?step.
50 ?component ssn:hasProperty ?tempProperty.
51 ?tempSensor sosa:observes ?tempProperty;
52 a brick:Temperature_Sensor;
53 td:name ?tempSensorIoT;
54 ref:hasTimeseriesID ?tempSensorID.}
55 GROUP BY ?tempSensorID ?component ?tempSensorIoT
56 ORDER BY ASC(?distance) LIMIT 1}
57 }

```

Listing 3: Query for data points and design properties of the radiator.

Table 4 displays the query results. The thermal power $\dot{Q}_{rad,n}$ of 1980 kW, supply water temperature $\theta_{rad,1,n}$ of 75 °C and return water temperature $\theta_{rad,2,n}$ of 65 °C, exponential coefficient n_{rad} of 1.3, under the nominal condition, and the actuatable property water value position $\dot{a}_{rad}$ as well as its data points (i.e., time series ID and IoT ID) are identified.

Using a similar approach, we query required properties of reheating coil and dampers, which are detailed in Appendix A. The manual efforts for interpreting the data points and system drawing can be considerably reduced in this way.

5.3.3. Querying the control procedure

One important goal is to increase the reusability of MPC algorithms and to simplify their deployment. As shown in Listing 4, we can query the deployed sub-algorithms for system identification, disturbance forecasts, as well as the OCP of the MPC controller. The MPC controller algorithm is thus fully identifiable in the RDF graph and can then be transferred and reused elsewhere, such as another room or building. The specific web service or function can be called and executed with

Table 5
Base URLs for the procedures involved in inst:Controller_A162.

Procedure	Base URL
Optimization	http://bosch/control-extension/optimization/room_energy_conservation
System identification	http://bosch/control-extension/SI/rc_dynamic
Forecasts	https://bosch/control-extension/forecasts/ambience
	https://bosch/control-extension/forecasts/occupants
	https://bosch/control-extension/reference/temperature

these unique identifiers, carrying out the closed-loop control. Furthermore, the same algorithms, e.g., forecasts on the disturbances, can be shared more flexibly among different MPC controllers. The result for the MPC controller in room A.162 is shown as Table 5.

```

1 SELECT DISTINCT ?space ?controller
2 ?ocpURL ?siURL (GROUP_CONCAT(DISTINCT ?forecast
3 -URL; separator="\n") as ?forecastsURL)
4 where
5 {
6   ?space a bot-ext:HVACSpace ;
7   props:hasRevitId '1500542';
8   bot:containsZone ?room.
9   ?controller a control:ModelPredictiveContro
10  -ller;
11   seas:optimizes ?space;
12   pep:implements ?si, ?forecasts, ?ocp.
13   ?si a ?siClass.
14   ?siClass rdfs:subClassOf control:SystemIden
15   -tification;
16   hctl:hasTarget ?siURL.
17   ?forecast a ?forecastClass.
18   ?forecastClass rdfs:subClassOf seas:Forecas
19   -ting;
20   hctl:hasTarget ?forecastURL.
21   ?ocp a ?ocpClass.
22   ?ocpClass rdfs:subClassOf control:OptimalCon
23   -trolProblem;
24   hctl:hasTarget ?ocpURL. }
25 GROUP BY ?space ?controller ?optimization ?si
26 ?ocpURL ?siURL

```

Listing 4: Query for the MPC procedure for the room A.162.

5.3.4. Retrieving data from the infrastructure

With data point IDs in both the time series database and the IoT platform available, we can access the historical measurements and set the optimal actuation of a component. We deploy the InfluxDB Python client library for retrieving data from Influx DB version 2.7, and the Azure IoT Python SDK to connect to the Azure IoT Hub and write a new actuation proposal.

Listing 5 shows a Flux query with a given database setting, a predefined time range, and a list of time series sensor IDs. Combined with the previously explained data point query, e.g., Listing 3, the historical measurements are returned in an automated way and can be sent to the data pre-processing and model training pipelines.

```

1 tag_filter_condition = ' or '
2 .join([f'r["{tag_name}"] == "{sensor_id}"'
3 for sensor_id in sensor_timeseries_ids])
4 query = f'''
5 from(bucket: "{bucket}")
6 |> range(start: {min_timestamp},
7 stop: {max_timestamp})
8 |> filter(fn: (r) =>
9   r["measurement"] == "{measurement}")
10 |> filter(fn: (r) => {tag_filter_condition})
11 '''

```

Listing 5: Flux query to retrieve historic data.

As for the real-time data from the IoT hub, a client is authorized to listen to all the device-to-cloud messages and filter out only the necessary data points based on the device's twin IDs. The stream measurements are associated with the corresponding properties based on the metadata. For the closed-loop control, the optimal actuation calculated is sent directly to the corresponding actuator via the cloud-to-device message as shown in Listing 6.

```

1 datapoint = {"Datapoint": iot_id ,
2 "Value": optimal_value}
3 client.send_message(json.dumps(datapoint))

```

Listing 6: Set the optimal actuation via the IoT hub.

5.3.5. Estimating heat flows and identifying the system dynamic for each thermal zone

Some heat flow measurements required in MPC configuration, especially the ones dealt in the system identification procedure, are frequently missing in real practice. Because the thermal meters are quite expensive and are not installed on every component in individual rooms. For instance, a typical hydraulic heating circuit nowadays has only one thermal meter, temperature sensor, and flow sensor for the supply branch and return branch, respectively. However, each radiator does not have temperature sensors, flow sensors, or a separate thermal meter. The heat flows, therefore, need to be either recalculated based on other available data, such as the measurements of flow sensors and temperature sensors or approximated using domain knowledge. This causes considerable manual effort.

In this section, we present how to automatically calculate the heat flow delivered from the HVAC system $\dot{Q}_{\text{HVAC}}$ into individual room, which is often required in identifying an RC model for the thermal envelope [18,50,91]. Then, based on the control procedure described in Section 5.3.3, we use the estimated heat flow to train the RC model that predicts the zone temperature in the MPC algorithm, and compare the prediction with measurements. The $\dot{Q}_{\text{HVAC}}$ consists of two parts: the heat flows delivered by VAV $\dot{Q}_{\text{vent}}$ and by the radiators $\dot{Q}_{\text{rad}}$, respectively. $\dot{Q}_{\text{vent}}$ is calculated using Eq. (2).

$$\dot{Q}_{\text{vent}} = \rho_a C_a \dot{V}_{z,1} (\vartheta_{z,2} - \vartheta_{z,1}) \quad (2a)$$

$$\dot{V}_{z,1} = f(\alpha_{\text{damp},1}, d_{\text{damp},1}) \quad (2b)$$

where ρ_a denotes the density of the air, and C_a stands for the mass-specific heat capacity of the air. The volume airflow rate has a linear correlation of the damper position $\alpha_{\text{damp},1}$ depending on the damper diameter $d_{\text{damp},1}$, as we use a self-regulating damper. The required metadata include damper property $d_{\text{damp},1}$ and time series ID of $\vartheta_{z,2}$, $\vartheta_{z,1}$ and $\alpha_{\text{damp},1}$.

The estimation of $\dot{Q}_{\text{rad}}$ are based on the thermal meter measurement of the entire heating circuit, as no further sensors of the individual radiators are available. We allocate the total heat flows of all radiators based on the heat load distribution among zones described by Stock et al. [51]. To simplify the problem, the heat transfer via the internal surfaces is neglected, because the temperature differences among zones are rather small in the office area. The thermal power of the radiators in the i th HVAC zone can be approximated using Eq. (3).

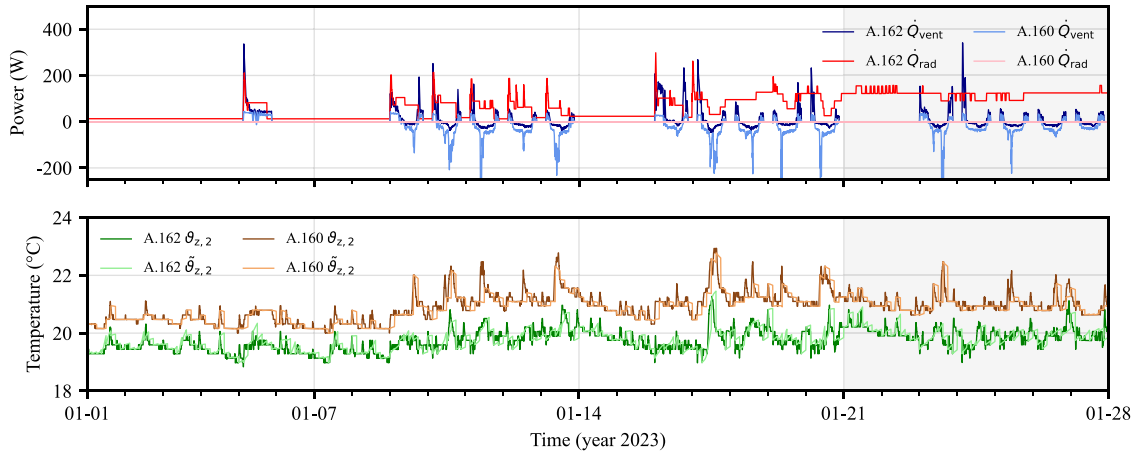


Fig. 12. Automated heat flows estimation and system identification of RC model for both rooms A.162 and A.160. For the RC model, the training period has a white background, and the testing period is in gray. When predicting $\vartheta_{z,2}$, the state is updated with real measurement $\vartheta_{z,2}$ every prediction horizon (3 h).

$$\dot{Q}_{rad,i} = \frac{\sum_{k \in \mathcal{M}_i} A_{i,k} U_{i,k} (\vartheta_{z,i} - \vartheta_{amb})}{\sum_{i=1}^{N_z} \left(\sum_{k \in \mathcal{M}_i} A_{i,k} U_{i,k} (\vartheta_{z,i} - \vartheta_{amb}) \right)} \dot{Q}_{rad,total} \quad (3)$$

in which ϑ_{amb} denotes the outside air temperature, $\mathcal{M}_i$ the set of indexes corresponding to the exterior surfaces in the i th zone, N_z the number of HVAC zones that the thermal meter $\dot{Q}_{rad,total}$ accounts for, and $A_{i,k}$ for the area and $U_{i,k}$ for the heat transfer coefficient of the k -th exterior surface. To perform the calculation, metadata including $A_{i,k}$, $U_{i,k}$, and boundary conditions of surfaces in the given thermal zone, and time series IDs for $\dot{Q}_{rad,total}$, $\vartheta_{z,2}$, ϑ_{amb} is mandatory.

The generated RDF graph contains all the information above, and can, therefore, be used to automate the heat flow estimation. Given a room ID, the metadata about the HVAC components and sensors' ID is firstly determined using the RDF graph as described in Section 5.3.2; With the metadata, the historical measurements are fetched as explained in Section 5.3.4. We performed the heat flow estimation on two rooms, A.162 and A.160, from January 1st to January 28th, 2023. Room A.160 is another meeting room, which has no exterior walls or radiators. For both rooms, we trained an RC model with the first 3 weeks' data, and validated it on the last week's data. Fig. 12 displays the results of heat flow estimation, and the predicted zone temperature $\vartheta_{z,2}$ and measured zone temperature $\vartheta_{z,2}$ for both rooms. Based on the estimated heat flows, in room A.162, the heating power is mainly provided by the radiator and sometimes also by the VAV box, as $\vartheta_{z,2}$ of room A.162 is most of the time below 21 °C. In room A.160, the heat flow of the radiator is constantly zero, as the zone is not directly exposed to the outside ambience and no radiator is installed in the room; the heat flow of the VAV box is sometimes negative, as $\vartheta_{z,2}$ of room A.160 sometimes rises higher 21 °C. The system identification procedure uses a training dataset with a length of 21 days and a sampling interval of 5 min, whose technical implementation is detailed in [51]. We use the RC model to predict 3-h-ahead zone temperature with a time interval of 5 min. The Root Mean Square Error (RMSE) between prediction $\hat{\vartheta}_{z,2}$ and real measurement $\vartheta_{z,2}$ of room A.162 and A.160 in the testing period are 0.28 °C and 0.27 °C, respectively. Our results about estimated heat flows corroborate with the real operation of the HVAC system, reflected by the measured room temperatures. The prediction performance of the RC models identified using the estimated thermal powers aligns with previous studies [90,94], confirming its validity as a system dynamic model in an MPC algorithm. Our method represents a viable approach based on the RDF graph to reduce sharply human efforts required in MPC configuration, where engineers nowadays spend considerable time to understand the HVAC systems and manually prepare the data.

6. Discussion

Because the main hindrance of wide adoption of MPC in the building domain is due to the data silos, Section 6.1 discusses how our study based on the semantic data modelling can make the MPC design more salable. Then, limitations of the current work, as well as the planned future research, are elaborated in Section 6.2.

6.1. Impact and contributions

A shareable machine-interpretable data model is crucial for reducing manual efforts in MPC design and enabling data exchange among heterogeneous systems. While semantic data modelling has proven effective in integrating diverse data in AEC/FM domains, previous studies lack thorough elaboration on metadata crucial for MPC implementation and its modelling of HVAC systems. This study uses semantic data modelling to address this engineering challenge in buildings. Academic contributions are discussed in Sections 6.1.1 and 6.1.2, and the technical contribution in Section 6.1.3.

6.1.1. Data model for MPC design

Previous studies have emphasized the semantics mainly of BMS data points and their connections to spaces or HVAC systems [47,49], and have suggested a model of rule-based control logic [35,48,49]. However, we believe that the metadata of the MPC algorithm, building material and topology, the HVAC components and system topology, and the associated sensors and actuators are all crucial. Because metadata of building and HVAC systems provide the detailed context of the systems, which can be used to select appropriate control model structure and parameters, locate the corresponding sensors and actuators, and parameterize the constraints subjected to room or components in the OCP formulation. Furthermore, we suggest modelling the MPC algorithm according to the three typical design procedures: system identification, disturbance forecast, and optimal control problem. Because control engineers are familiar with such concepts. We have reviewed the state-of-the-art ontologies in the four domains mentioned above in Section 2.3 and managed to design the T-box completely aligning to the well-established openly available ontologies, as explained in Section 3. Our modelling approach sheds light on how to model the MPC application in buildings using existing standard schema, as recommended in the building and energy modelling community [25]. Described via ontologies, MPC algorithms with such interfaces are transferable from building to building.

6.1.2. Graph-based data preparation

For MPC design, we demonstrated querying metadata for control configuration using SPARQL and interfacing with existing BMS infrastructure. We estimated heat flows delivered by HVAC systems into thermal zones using domain knowledge encoded in the RDF graph, then used this estimation to identify the RC model of the thermal zone. Experiments on two meeting room types show RMSEs of 3h-ahead temperature prediction below 0.3 °C, suitable for MPC design. This semantic-graph-based workflow reduces manual efforts in system understanding, data collection, and control model parameterization. Results demonstrate improved zone-level reusability explicitly. Provided with valid RDF graphs, our approach can ideally be adopted to determine RC models for thermal zones in any buildings with VAV and radiator heating systems.

6.1.3. Improvement to the graph generator

To enable practical adoption, our work also made a technical contribution by extending a graph generator for scalable utility in actual buildings. Compared with the original graph generator, Revit2RDF now extracts detailed geometry of thermal zones, actuator and sensor metadata, HVAC component properties, and functional information of HVAC systems. It also includes validation in the graph generation process for interactive error fixing.

6.2. Limitations and future work

Our work has some limitations that must be considered. First, the utility of the proposed data model in the close-loop control needs to be further investigated. Although the T-box has considered the modelling of stream data and resources for algorithm deployment, the MPC algorithm's concrete implementation is, unfortunately, not the main focus of this study. The integrity of the real-time data, such as the input measurements required by the specific algorithm, must be verified. Also, how to use the metadata of the resources, such as the service's endpoints, to simplify the algorithm's setup and deployment needs to be clarified. Second, this study demonstrated only how to perform graph-based system identification for the RC model. We selected the RC model in the case study because it is the most popular dynamic model of the building envelope. The direct utility of the proposed method for another kind of dynamic model, such as an artificial neuron network, could be restricted because the dynamic models have disparate input features. Nevertheless, using metadata from the semantic graph to select informed model parameters and structures is beneficial for other system identification algorithms. Third, we developed the RDF graph generator based on the proprietary software Revit and only tested it in version 2022. Minor modifications of the graph generator can be required if another version of Revit is used for the BIM project. As we have made the generator code open-source, we hope our research will serve as a base for the community to strengthen the graph generator, and develop further tools that support other BIM environments.

To further our research, we will investigate how semantic data modelling contributes to the MPC deployment and its real-time closed-loop control performance. We believe deploying the MPC as a web service is a promising approach, as suggested in [95]. Further studies that use semantic graphs to configure other types of control-oriented models, e.g., white- and black-box models, will need to be performed so that we understand how metadata may benefit different modelling approaches and shall be used in different smart building applications.

7. Conclusion

Previous studies have presented promising potential for improving energy systems' efficiency using MPC. Nevertheless, most of the developed MPC algorithms are tailored for specific buildings, and are difficult to be directly transferred to other buildings. This is caused by the cumbersome data exchange among the diverse data sources.

This paper has demonstrated the critical role of a shareable machine-interpretable data schema for the scalable MPC design of a building and its HVAC systems. The absence of such a model has formerly led to considerable manual efforts in understanding and collecting the system data required for MPC configuration and deployment. Prior investigations have emphasized the semantics of BMS data points and their connections to space or HVAC systems. However, we believe that metadata for the MPC algorithm, building physics and geometry, HVAC system topology, and associated sensors and actuators is significant because this information provides detailed context for configuring appropriate control-oriented models and parameterizing OCP formulations. By leveraging established openly available ontologies such as SEAS, FSO, TSO, MEP, BOT, BEO, PROPS, SOSA/SSN, Brick, TD, and HCTL, we proposed a comprehensive data model that integrates key metadata across the aforementioned domains. Based on BIM, a software tool has been built to generate RDF graphs adhering to the proposed schema for real buildings. Using the developed software, the proposed data model was demonstrated by example instance graphs generated from the Revit model of a real office building. The instance graphs were further evaluated against multiple use cases, where crucial inputs for the MPC algorithms, such as metadata, sensor measurements, estimation of heat flows, and system dynamic model, are automatically determined. Our work underscores the potential of semantic data modelling to streamline and standardize MPC configuration in the building sector, offering valuable insights and tools for researchers and developers aiming to enhance predictive control systems in the building industry. We envision future research investigating how semantic data benefits different types of control-oriented models, the data integrity, and the performance of the MPC algorithm in a closed-loop manner.

CRediT authorship contribution statement

Lu Wan: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Ferdinand Rossa:** Writing – original draft, Software, Data curation. **Torsten Welfonder:** Writing – review & editing, Supervision, Methodology. **Ekaterina Petrova:** Writing – review & editing, Supervision, Methodology. **Pieter Pauwels:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Data schema, dataset and code

The proposed data model (T-box, extension ontologies), instance graph of the case study building (A-box), SHACL shapes, and the SPARQL queries mentioned are available at <https://github.com/boschresearch/semantics4mpc>. The graph generator Revit2RDF is available at <https://github.com/boschresearch/revit2rdf>.

Appendix B. Detailed view of HVAC system

Fig. B.13 presents all the HVAC components and sensors that serve room A.162.

Data availability

The data schema, dataset as well as the code files are hosted on Github.

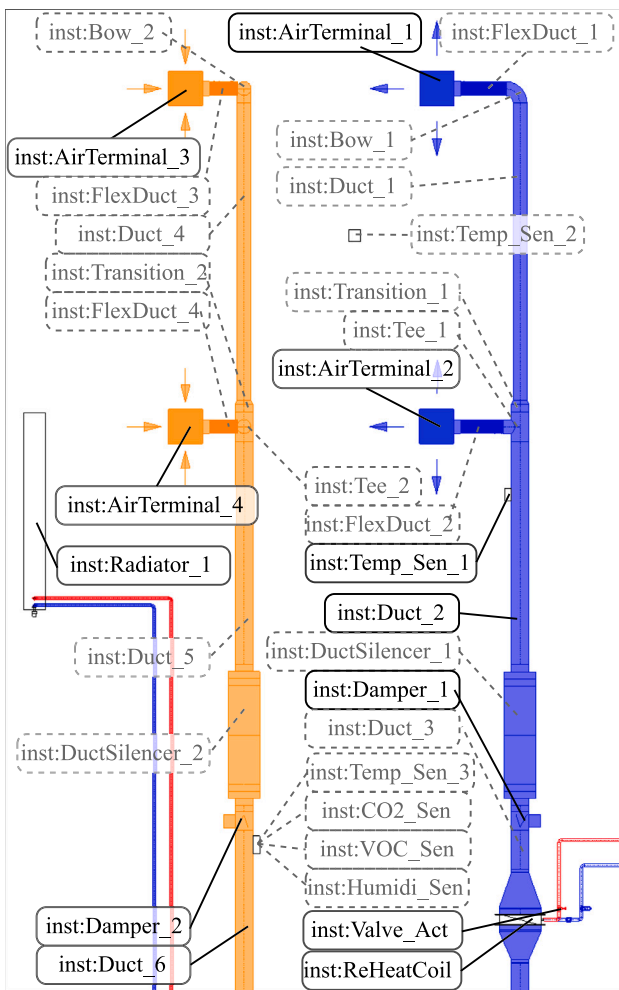


Fig. B.13. 2D illustration of all the HVAC systems and sensors instances in room A.162. The black instances are depicted in the A-box figure, while the grey ones are omitted.

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